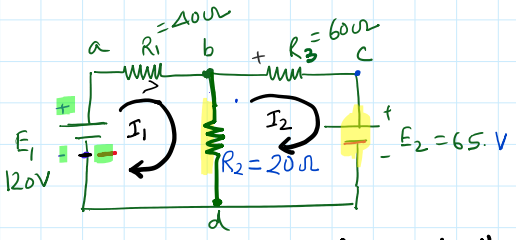
Example 2

Network Analysis by Mesh Currents



KVL: Mesh ①

$$-E_1 + R_1 I_1 + R_2 (I_1 - I_2) = 0$$

$$(R_1 + R_2) I_1 - R_2 I_2 = E_1 \quad \text{—— ①}$$

In the circuit:
$j = 2$, $b = 3$
$m = b - (j-1) = 3 - (2-1)$
$= 2$ (two independent meshes)

KVL: mesh ②

$$R_3 I_2 + E_2 + R_2 (I_2 - I_1) = 0$$

$$-R_2 I_1 + (R_2 + R_3) I_2 = -E_2 \quad \text{—— ②}$$

$$\left.\begin{aligned} (R_1+R_2) I_1 - R_2 I_2 &= E_1 \\ -R_2 I_1 + (R_2+R_3) I_2 &= -E_2 \end{aligned}\right\} \quad I_1 = ?, \; I_2 = ?$$

Using Substitution Method:

$$60 I_1 - 20 I_2 = 120 \quad \text{---- (1)'} \Rightarrow I_2 = 3I_1 - 6$$

$$-20 I_1 + 80 I_2 = -65 \quad \text{---- (2)'}$$

$$-20 I_1 + 8(3I_1 - 6) = -65$$

$$\Rightarrow I_1 = 1.086 \text{ A}$$

and $I_2 = -0.341$ A

Matrix Method:

$$\left.\begin{aligned} R_{11} I_1 + R_{12} I_2 &= E_1 \\ R_{21} I_1 + R_{22} I_2 &= -E_2 \end{aligned}\right\} \quad \begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} E_1 \\ -E_2 \end{bmatrix}$$

$R_{11} = R_1 + R_2 =$ Self resistance of mesh ①

$$R_{ii} = \sum_{j=1}^{K} r_{ij} \Rightarrow$$

$$R_{ij} = -r_{ij}$$

Similarly $R_{22} = r_{21} + r_{22} + r_{23} +$

$R_{ii} = r_{i1} + r_{i2} + \cdots + r_{iK}$

This is called self-loop resistance.

$$R_{11} = r_{11} + r_{12} + \cdots + r_{1K}.$$

Now we define mutual resistance R_{ij} as following.

$$R_{ij} = -r_{ij} \Rightarrow R_{12} = -r_{12}$$

$$[I] = [R]^{-1}[V] = \frac{1}{\Delta R} \begin{bmatrix} R_{C11} & R_{C12} \\ R_{C21} & R_{C22} \end{bmatrix}^T$$

$$[R] = \begin{bmatrix} R_1+R_2 & -R_2 \\ -R_2 & R_2+R_3 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} = \begin{bmatrix} 60 & -20 \\ -20 & 80 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \frac{1}{\Delta R} \begin{bmatrix} R_{22} & -R_{21} \\ -R_{12} & R_{11} \end{bmatrix}^T \begin{bmatrix} E_1 \\ -E_2 \end{bmatrix} = \frac{1}{\Delta R} \begin{bmatrix} R_{22} & -R_{12} \\ -R_{21} & R_{11} \end{bmatrix} \begin{bmatrix} E_1 \\ -E_2 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \frac{1}{\Delta R}\begin{bmatrix} R_{22} & R_{12} \\ -R_{12} & R_{11} \end{bmatrix}\begin{bmatrix} E_1 \\ -E_2 \end{bmatrix} = \frac{1}{\Delta R}\begin{bmatrix} R_{22}E_1 + R_{12}E_2 \\ -R_{12}E_1 + R_{11}E_2 \end{bmatrix}$$

$$I_1 = \frac{R_{22}E_1 + R_{12}E_2}{\Delta R} = \frac{\begin{vmatrix} E_1 & R_{12} \\ -E_2 & R_{22} \end{vmatrix}}{\Delta R} = 1.86A$$

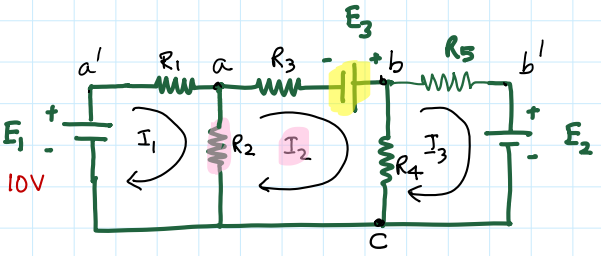
$$I_2 = \frac{\begin{vmatrix} R_{11} & E_1 \\ R_{21} & -E_2 \end{vmatrix}}{\Delta R} = -0.34A$$

$$I_{R2} = I_1 - I_2 = 1.86 - (-0.34) = 2.2A$$

Example 3:

Let us take another example:

$E_1 = 10V$ $\quad E_3 = 15V$
$R_1 = 10\Omega$ $\quad R_5 = 5\Omega$
$R_2 = 5\Omega$ $\quad E_2 = 10V$
$R_3 = 15\Omega$ $\quad R_4 = 10\Omega$



$m = b - (j-1) = 5 - (3-1) = 3$

Mesh ①

$$-E_1 + R_1 I_1 + R_2(I_1 - I_2) = 0$$

$$(R_1+R_2)I_1 - R_2 I_2 = E_1 \quad \cdots\cdots\cdots ①$$

$$15I_1 - 5I_2 = 10 \quad \cdots\cdots\cdots (1)'$$

Mesh ②:

$$R_3 I_2 + (-E_3) + R_4(I_2 - I_3) + R_2(I_2 - I_1) = 0$$

$$-R_2 I_1 + (R_2+R_3+R_4)I_2 - R_4 I_3 = E_3 \quad \cdots\cdots ②$$

$$-5I_1 + 30I_2 - 10I_3 = 15 \quad \cdots\cdots\cdots (2)'$$

Mesh ③:

$$R_5 I_3 + E_2 + R_4(I_3 - I_2) = 0$$

$$-R_4 I_2 + (R_4+R_5)I_3 = -E_2 \quad \cdots\cdots ③$$

$$-10I_2 + 15I_3 = -10 \quad \cdots\cdots\cdots (3)'$$

A. Substitution Method:

We can solve (1)' – (3)' using substitution method.

For example from (1)', $I_1 = \frac{10 + 5I_2}{15} = \frac{2 + I_2}{3}$

For example from (1)', $I_1 = \frac{10+5I_2}{15} = \frac{2+I_2}{3}$

From (3)' $I_3 = \frac{-10+10I_2}{15} = \frac{-2+2I_2}{3}$

. Substituting in (2)',

$$-5\left(\frac{2+I_2}{3}\right) + 30I_2 - 10\times\left(\frac{-2+2I_2}{3}\right) = 15$$

$$\frac{-10-5I_2+20-20I_2}{3} + 30I_2 = 15$$

$$-25I_2 + 10 + 90I_2 = 45$$

$$13 \cancel{65} I_2 = \cancel{35}\ 7 \Big\} \quad I_2 = \frac{7}{13} = 0.538A$$

$I_1 = \frac{2+I_2}{3} = 0.846A$

$I_2 = \frac{7}{13} = 0.538A$

$I_3 = \frac{-2+2I_2}{3} = -0.308A$

B. Using Matrix Method : Crammer's Rule

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{33} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} E_1 \\ E_3 \\ -E_2 \end{bmatrix}$$

$$[R][I] = [V]$$

$$[I] = [R]^{-1}[V]$$

$$R_{11} = R_1 + R_2$$
$$R_{12} = R_{21} = -R_2$$
$$R_{13} = 0$$
$$R_{21} = -R_2$$
$$R_{22} = R_2 + R_3 + R_4$$
$$R_{23} = -R_4$$
$$R_{31} = 0$$
$$R_{32} = -R_4$$
$$R_{33} = R_4 + R_4$$

$$[R] = \begin{bmatrix} 15 & -5 & 0 \\ -5 & 30 & -10 \\ 0 & -10 & 15 \end{bmatrix}_{3\times 3}$$

$$[I] = [R]^{-1}[V]$$

Using Cramer's Rule:

$$I_1 = \frac{1}{\Delta R}\begin{vmatrix} E_1 & R_{12} & R_{13} \\ E_3 & R_{22} & R_{23} \\ -E_2 & R_{32} & R_{33} \end{vmatrix} = \frac{1}{4875}\begin{vmatrix} 10 & -5 & 0 \\ 15 & 30 & -10 \\ -10 & -10 & 15 \end{vmatrix} = 0.8462\,A$$

$$I_2 = \frac{1}{\Delta R}\begin{vmatrix} R_{11} & E_1 & R_{13} \\ R_{21} & E_3 & R_{23} \\ R_{31} & -E_2 & R_{33} \end{vmatrix} = \frac{1}{4875}\begin{vmatrix} 15 & 10 & 0 \\ -5 & 15 & -10 \\ 0 & -10 & 15 \end{vmatrix} = 0.5385\,A$$

$$I_3 = \frac{1}{\Delta R}\begin{vmatrix} R_{11} & R_{12} & E_1 \\ R_{21} & R_{22} & E_3 \\ R_{31} & R_{32} & -E_2 \end{vmatrix} = \frac{1}{4875}\begin{vmatrix} 15 & -5 & 10 \\ -5 & 30 & 15 \\ 0 & -10 & -10 \end{vmatrix} = -0.3077\,A$$

C. Matrix Method: By computing Inverse

Another way is that we can directly find the inverse of matrix [R]; $[R]^{-1}$ =

$$\begin{bmatrix} 0.0718 & 0.0154 & 0.0103 \\ 0.0154 & 0.0462 & 0.0308 \\ 0.0103 & 0.0308 & 0.0872 \end{bmatrix}$$

Then $I_1 = R_{inv}(1,:)\begin{bmatrix} E_1 \\ E_3 \\ -E_2 \end{bmatrix} = 0.0718\times 10 + (0.0154\times 15) + 0.0103\times(-10)$

$= 0.8462\,A$

Similarly, $I_2 = R_{inv}(2,:) * [10 \;\; 15 \;\; -10]^T = 0.5385\,A$

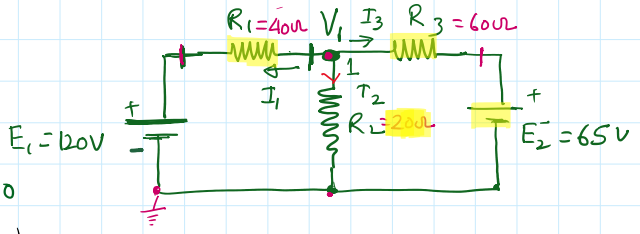
$I_3 = R_{inv}(3,:) * [10 \;\; 15 \;\; -10]^T = -0.3077\,A$

Nodal Analysis: Circuit Analysis Using Node-pair Equation.

KCL = Kirchhoff's Current Law: $\sum_{j=1}^{k} I_j = 0$

Nodal variables = j − 1

Example 4:



KCL at node ①

$$\frac{V_1 - E_1}{R_1} + \frac{V_1 - 0}{R_2} + \frac{V_1 - E_2}{R_3} = 0$$

$$\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) V_1 = \frac{E_1}{R_1} + \frac{E_2}{R_3}$$

$$V_1 = \frac{E_1/R_1 + E_2/R_3}{\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)} \Rightarrow V_1 = 44.54\,V$$

n = No. of node-pair current equations = j − 1

$$[G][V] = [I]$$

(G: conductance)

$$I_1 = \frac{V_1 - E_1}{R_1} = -1.88\,A$$

$$I_2 = \frac{V_1}{R_2} = \frac{44.54}{20} = 2.23\,A$$

$$I_3 = \frac{V_1 - E_2}{R_3} = \frac{44.54 - 65}{60} = -0.34\,A$$

Now let us consider the following circuit:

Find the currents in R_1, R_2, R_3, R_4, R_5 using node-pair method.

$J = 3$

n = No of unknown variables $= J - 1 = 2$

